Calibrating Learners' Trust in Generative AI: A Narrative Review of Design Interventions for Appropriate Reliance in Education

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Abstract

Background. Students now use generative AI (GenAI) to explain concepts, check answers, write code, and draft essays. Whether that helps or hurts them depends less on how much they trust these systems than on whether their trust is calibrated: relying on AI when it is right, questioning it when it is wrong, and not using it at all when doing the work yourself is the point of the exercise. Human-computer interaction (HCI) research has produced many interventions meant to encourage appropriate reliance on AI, but most were tested on decision-making or question-answering tasks, not on learning.

Aim. In this narrative review I ask which design interventions calibrate learners' trust in and reliance on GenAI, which outcomes they have actually been shown to change, and whether their benefits reach all learners equally.

Approach. I identified literature through Crossref, OpenAlex, PubMed, and arXiv searches, plus targeted searches for frequently cited studies on trust calibration and GenAI in education. I cite only peer-reviewed sources. The review is narrative, not systematic, and I describe the search so readers can judge its coverage.

Findings. Evidence from high schools and universities shows poorly calibrated reliance. Among 432 undergraduates, higher trust went with worse discrimination between correct and incorrect AI suggestions, and in a field experiment with high school students, unrestricted AI access during practice hurt later performance. GenAI brings its own difficulties: fluent and lengthy answers inflate confidence, models rarely express uncertainty and are overconfident when they do, and trust recovers slowly after confident errors. I group the interventions into four families: communicating reliability (uncertainty expressions, calibrated explanations, sources), forcing cognitive engagement (cognitive forcing, questioning, guarded tutors, metacognitive prompts), changing the costs and benefits of verification, and building learner capacity (AI literacy). Explanations on their own tend to increase reliance whether the AI is right or wrong. Interventions that make people think before accepting an answer reduce overreliance more reliably, but people like them less, and several studies find they help learners with higher need for cognition, prior knowledge, or AI literacy more. AI literacy moderates the link between trust and appropriate reliance, yet few AI literacy programs check what students actually learned.

Conclusions. I propose that appropriate reliance in education has two reference points, the AI's reliability and the learner's learning goal (Figure 1), and that most interventions address only the first. Progress will need outcome measures that separate trust level from calibration and from learning, evaluation with AI access removed, and routine checks for intervention-generated inequalities.

Keywords: trust calibration; appropriate reliance; overreliance; generative AI; education; metacognition; human-AI interaction; equity

1. Introduction

For a long time, trust has been treated as the variable that governs how people use automation. Classic work separated misuse, overreliance that leads to failures of monitoring, from disuse, the neglect or underuse of automation (Parasuraman & Riley, 1997). Lee and See (2004) argued that because fully understanding an automated system is often impractical, trust guides reliance, and they framed the design problem as one of appropriate reliance rather than more reliance. Later syntheses sorted the many factors that shape trust into dispositional, situational, and learned layers (Hoff & Bashir, 2015) and distinguished warranted trust, grounded in a system's actual trustworthiness, from unwarranted trust (Jacovi et al., 2021).

GenAI has brought this question into classrooms. A student can now ask a large language model (LLM) to solve a problem, explain an error, or summarize a reading, and the model will answer fluently whether or not it is right. The evidence on whether this helps learning is honestly mixed. In a randomized trial, college students learned more in less time with a carefully designed AI tutor than in an active-learning class (Kestin et al., 2025). ChatGPT-generated hints in mathematics produced learning gains with no statistically significant difference from human-authored hints, even though the ChatGPT help initially failed quality checks on 32% of problems (Pardos & Bhandari, 2024). Yet in a field experiment with nearly a thousand high school students, unrestricted access to GPT-4 during practice improved performance while access lasted and left students doing worse than controls once it was taken away (Bastani et al., 2025). Students themselves are of two minds. In a survey of 399 university students in Hong Kong, attitudes toward GenAI were generally positive, but students also worried about accuracy, privacy, and the effect on their personal development (Chan & Hu, 2023).

I read these results as pointing to the same issue. Whether AI helps a student learn depends on how the student relies on it, and that depends on trust that may or may not be warranted. Several recent reviews cover parts of this. Mehrotra et al. (2024) systematically reviewed approaches for fostering appropriate trust in human-AI interaction, including confidence scores, explanations, and uncertainty communication. Wischnewski et al. (2023) reviewed 96 publications on trust calibration interventions for automated systems and concluded that the way calibration is measured often limits what can be said about the interventions. In education, Zhai et al. (2024) reviewed 14 studies on how overreliance on AI dialogue systems affects students' cognitive abilities. What I could not find was a synthesis that asks, specifically for learners using GenAI, which interventions have been tested, what they actually change (trust level, how appropriate reliance is, or learning itself), and for whom they work.

This review tries to fill that gap. I have three aims:

1. To clarify what "appropriate reliance" should mean when the user is a learner, drawing on the human-automation trust tradition and recent work on GenAI and metacognition.
2. To synthesize the evidence on design interventions meant to calibrate trust and reliance, keeping evidence from learning contexts separate from evidence gathered in general decision tasks.
3. To examine whether intervention effects differ across learners, and to propose a research agenda that makes equitable calibration an explicit design goal.

2. Approach to the Literature

This is a narrative review. My goal is to interpret and connect a body of work, not to estimate pooled effects. Following guidance that narrative reviews should report how literature was identified (Baethge et al., 2019), I summarize the search here.

Sources and dates. I ran searches in September 2026 in Crossref, OpenAlex, and PubMed, and searched arXiv to find peer-reviewed versions of recent work. Search terms combined concepts of trust and reliance ("trust," "trust calibration," "appropriate reliance," "overreliance"), AI ("generative AI," "large language model," "ChatGPT," "AI assistant"), learning ("students," "education," "learning," "novice programmers"), and intervention ("explanation," "uncertainty," "cognitive forcing," "metacognitive prompts," "AI literacy"). I added targeted title searches for frequently cited studies in the trust calibration and AI-in-education literatures.

Selection. I included studies that (a) measured trust in or reliance on AI among learners, (b) tested an intervention meant to change reliance on AI or engagement with AI output, in any population, where the mechanism seemed relevant to learning, or (c) measured learning or skill outcomes after AI-assisted work. I also included conceptual papers where they defined terms used across the literature.

Limitations of the approach. The search was not exhaustive and covered only English-language sources. I excluded preprints and working papers, so some recent findings that are not yet peer reviewed are missing. I report findings as the original authors state them, and where I draw my own inferences, I say so.

3. What Counts as Appropriate Reliance for a Learner?

3.1 Trust, reliance, and calibration

Three terms need to be kept apart. Trust is an attitude, a belief or expectation about a system. Reliance is a behavior: accepting or acting on the system's output. Calibration describes how well the two fit what the system can actually do; for reliance to be appropriate, users' trust should reflect the system's capabilities (Wischnewski et al., 2023). This matters because many studies measure only one of the three. A survey of empirical methods for studying trust in AI-assisted decision-making found that definitions of trust often were not built into experimental protocols, which makes findings hard to compare (Vereschak et al., 2021). Schemmer et al. (2023) proposed Appropriateness of Reliance as a quantifiable, two-dimensional measure, since people need both to rely on correct AI advice and to turn down incorrect advice. Mehrotra et al. (2024) likewise found no single agreed definition of appropriate trust and proposed mapping related concepts across beliefs, intentions, and actions.

For GenAI one more distinction helps. Steyvers et al. (2025) separate the calibration gap, the difference between people's confidence in an LLM's answers and the model's own confidence, from the discrimination gap, how well people can tell correct from incorrect answers. A learner can be well calibrated on average and still be unable to tell individual right and wrong answers apart, and those two problems call for different fixes.

3.2 Why learning adds a second reference point

In decision-support research, reliance is usually judged against one standard: was the AI correct? Learning complicates that. The point of a practice problem is not to produce a correct answer but to build the ability to produce correct answers later. Cognitive offloading is defined as "the use of physical action

to alter the information processing requirements of a task so as to reduce cognitive demand" (Risko & Gilbert, 2016). In a learning task, the cognitive demand is often exactly what is being practiced.

So I propose that appropriate reliance for learners has two reference points. The first is the AI's reliability: accepting correct output and rejecting incorrect output. The second is the learning goal: relying on AI for parts of a task that are not the learning target, and doing the targeted work yourself. A student who accepts a correct AI solution to a practice problem is relying appropriately by the first standard and inappropriately by the second. Bastani et al. (2025) show the difference clearly. Students with unrestricted GPT-4 access did better with the tool and worse afterward, using it as what the authors call a "crutch."

This two-reference-point framing is my own synthesis, not an established construct. I offer it because most of the interventions reviewed below were designed around only the first reference point. Figure 1 lays out the four situations that result.

Figure 1. Two reference points for appropriate reliance in learning (proposed here).

AI output	Component is not the learning target	Component is the learning target
Correct	Rely on the AI. Appropriate by both standards.	Attempt first, then compare with the AI. Relying would be accurate but would skip the practice.
Incorrect	Verify and reject. Requires the learner to detect the error.	Attempt first, then detect and reject. The error becomes a learning opportunity only if it is caught.

Note. Decision-support research judges reliance only by rows (was the AI correct?). Learning adds the columns. My reading is that the learning losses from unrestricted AI use during practice reported by Bastani et al. (2025) come mostly from the right-hand column.

3.3 Why generative AI makes calibration harder

Several properties of current LLM-based systems make calibrated trust harder to reach than it was with earlier automation.

Fluency without calibrated confidence. Users overestimate the accuracy of LLM answers, and longer explanations raise their confidence even when the extra length adds no accuracy (Steyvers et al., 2025). The models themselves rarely signal doubt. An analysis of publicly deployed language models found they were reluctant to express uncertainty even when their answers were wrong; when prompted to state their confidence, they tended to be overconfident, with an average error rate of 47% among confident responses (Zhou et al., 2024). In the same work, users relied heavily on model outputs whether or not certainty was marked, and the human preference data used to fine-tune models showed a bias against text that expressed uncertainty. The signal that calibration depends on is often missing at the source.

Beliefs and affect shape trust apart from reliability. Trust in AI has cognitive and emotional parts. A review of empirical research linked tangibility, transparency, reliability, and immediacy behaviors to cognitive trust, and anthropomorphism specifically to emotional trust (Glikson & Woolley, 2020). How a system is introduced matters too. In one experiment, participants used the same conversational AI after

being told it had caring, manipulative, or no motives. Those who perceived a caring motive also saw the AI as more trustworthy, empathetic, and better-performing, and the priming effects were stronger for a more sophisticated model (Pataranutaporn et al., 2023). My reading is that tutors designed to feel warm and supportive may raise trust for reasons that have nothing to do with their accuracy.

Trust changes over time, and errors cost more than they should. In a betting game, a few incorrect predictions made with inaccurate confidence damaged user trust and performance, and trust recovered very slowly. Confidently incorrect and unconfidently correct outputs hurt trust in different ways (Dhuliawala et al., 2023). Over a semester, a student's trust is shaped by a history of running into the AI's errors, which single-session studies cannot capture.

4. How Well Calibrated Is Learners' Reliance?

Trust predicts worse discrimination. In a study of 432 undergraduates working on Python output-prediction problems with an AI assistant that gave both accurate and deliberately misleading suggestions, higher trust was associated with lower appropriate reliance, meaning weaker discrimination between correct and incorrect suggestions (Pitts et al., 2026b). An earlier study by the same group found that appropriate reliance was related to programming self-efficacy, programming literacy, and need for cognition, while overreliance correlated with post-task trust and satisfaction (Pitts et al., 2026a). Survey research with university computer science students found varying levels of trust in GenAI, along with different levels of confidence and motivation (Amoozadeh et al., 2024).

Fluency inflates confidence. In experiments with LLM answers to multiple-choice and short-answer questions, users overestimated the accuracy of responses that came with default explanations, and longer explanations raised confidence even when the extra length did not improve accuracy (Steyvers et al., 2025). In fact-checking, users reading LLM explanations were more efficient than those using a search engine but overrelied on the LLM when its explanation was wrong (Si et al., 2024).

Reliance can create an illusion of competence. In observational lab sessions with novice programmers, students who were not struggling used GenAI to speed up work they already understood and ignored unhelpful suggestions. Struggling students, by contrast, had their metacognitive difficulties compounded, overestimated their performance, and finished with what the authors call an illusion of competence (Prather et al., 2024). In a survey of 319 knowledge workers, higher confidence in GenAI was associated with less critical thinking, while higher self-confidence was associated with more (Lee et al., 2025). That sample was not students, but the pattern fits the student findings.

Reliance can displace learning. In a randomized experiment with 1,625 students across 10 courses, students tended to rely on AI prompts for writing peer feedback rather than learn from them. When the AI was removed, self-monitoring checklists partly filled the gap but were less effective than the AI assistance itself (Darvishi et al., 2024). University students using ChatGPT for a writing task improved their essay scores more than comparison groups, but their knowledge gain and transfer did not differ, and the authors warn of "metacognitive laziness" (Fan et al., 2025). Students researching a socio-scientific issue with ChatGPT had lower cognitive load than those using a search engine but produced lower-quality reasoning (Stadler et al., 2024). Survey evidence links frequent AI tool use with lower critical thinking scores, mediated by cognitive offloading (Gerlich, 2025). That evidence is correlational, so it cannot show that AI use caused the difference.

Well-designed AI can still support learning. The same literature has strong positive results when tools are built around pedagogy: the AI tutor designed around the same pedagogical practices as in-class active learning (Kestin et al., 2025), and the guarded "GPT Tutor" condition that largely removed the negative learning effects of unrestricted access (Bastani et al., 2025). So the question is not whether students should use GenAI. It is which designs calibrate their reliance.

5. Interventions: What Has Been Tried and What It Changes

I group the interventions in this literature into four families by mechanism. Table 1 lists representative studies, whether each was tested with learners, and which outcome it changed.

Table 1. Representative interventions for calibrating reliance on AI, by mechanism.

Mechanism	Intervention (study)	Population and task	Main reported effect
Communicate reliability	First-person uncertainty expressions (Kim et al., 2024)	Medical questions with LLM-infused search (N = 404)	Lower agreement with the system and higher accuracy; reduced overreliance
Communicate reliability	Explanations adjusted to model confidence (Steyvers et al., 2025)	Multiple-choice and short-answer questions	Narrowed calibration and discrimination gaps
Communicate reliability	Sources; inconsistent explanations (Kim et al., 2025)	Objective questions with an LLM-infused application (N = 308)	Less reliance on incorrect answers; explanations alone increased reliance on both correct and incorrect answers
Force engagement	Cognitive forcing functions (Buçinca et al., 2021)	AI-assisted decision task (N = 199)	Reduced overreliance vs. simple explanations; lowest subjective ratings; benefited high need-for-cognition users more
Force engagement	AI-framed questioning (Danry et al., 2023)	Judging logical validity of statements (N = 204)	Improved discernment of flawed statements vs. causal explanations
Force engagement	Guarded GPT tutor (Bastani et al., 2025)	High school math students	Largely mitigated learning losses after access was removed
Force engagement	Step-by-step guided engagement with AI code (Kazemitabaar et al., 2025)	Novice programmers	Most effective of the techniques tested, judged on friction, transfer without AI, and alignment of perceived

Mechanism	Intervention (study)	Population and task	Main reported effect
			with actual ability
Force engagement	Attempt before seeing LLM explanation (Kumar et al., 2025)	1,200 participants, math problems	Largest learning benefit when problems were attempted first
Force engagement	Metacognitive prompts in GenAI search (Singh et al., 2025)	University students	More active engagement and deeper inquiry; varied with metacognitive flexibility
Force engagement	Contrastive explanations anticipating misconceptions (Buçinca et al., 2025)	Decision task (N = 628)	Improved independent decision-making skill without lowering accuracy
Build capacity	Self-monitoring checklists (Darvishi et al., 2024)	University students, peer feedback	Partly filled the gap when AI was removed; less effective than AI assistance
Build capacity	Priming beliefs about the AI's motives (Pataranutaporn et al., 2023)	Interaction with a conversational AI	Perceiving a caring motive went with higher perceived trustworthiness, empathy, and performance of the same system

5.1 Communicating reliability

The most intuitive family tells users how reliable the AI is. The evidence says this can work, but the form matters. In a pre-registered experiment, first-person expressions of uncertainty ("I'm not sure, but…") reduced agreement with an LLM-infused search engine and increased accuracy, apparently by reducing overreliance on incorrect answers. Uncertainty phrased impersonally had weaker effects that were not significant (Kim et al., 2024). Showing confidence scores helped calibrate trust in an earlier decision-support study, but calibration alone did not improve AI-assisted decisions, which may also depend on whether people bring knowledge that complements the AI (Zhang et al., 2020). Adjusting LLM explanations to reflect the model's internal confidence narrowed both the calibration gap and the discrimination gap (Steyvers et al., 2025).

Explanations deserve particular caution. In user studies on three datasets, explanations raised the chance that people accepted AI recommendations regardless of whether they were correct (Bansal et al., 2021). In a recent LLM experiment, explanations increased reliance on both correct and incorrect answers (Kim et al., 2025), though the same experiment found less reliance on incorrect answers when sources were provided or when explanations contained inconsistencies. Presenting contrastive information, arguments for why a claim is both true and false, reduced overreliance in fact-checking (Si et al., 2024). For code,

highlighting the tokens most likely to be edited helped programmers, while highlighting tokens by generation probability did not (Vasconcelos et al., 2024). Exposing a model's raw uncertainty is not the same as communicating useful uncertainty.

Assessment. This is the best-studied family, but almost all of the evidence comes from decision-making or question-answering tasks rather than learning tasks. None of the reliability-communication studies in Table 1 measured learning with the AI removed.

5.2 Forcing cognitive engagement

The second family tries to make people think before they accept. Drawing on dual-process theory, Buçinca et al. (2021) argued that people rarely engage analytically with each AI recommendation and instead develop heuristics about when to follow it. Their cognitive forcing designs, which required people to engage with the task before or while seeing AI output, reduced overreliance compared with simple explainable AI approaches. Two findings from that study come up again and again in this review: participants rated the designs that reduced overreliance most as the least favorable, and the interventions helped people with higher need for cognition more.

Related designs change the AI's role. AI-framed questioning, which turns information relevant to an AI judgment into questions for the user, improved people's discernment of logically flawed statements compared with causal explanations, even from an always-correct system (Danry et al., 2023). Contrastive explanations that anticipate a user's likely misconception improved people's independent decision-making skills without lowering decision accuracy (Buçinca et al., 2025). It is one of the few studies in the general literature that measures skill development directly.

In education, this family has the strongest evidence available. The guarded "GPT Tutor" in Bastani et al. (2025), a randomized trial at a high school in Turkey, largely removed the learning losses seen with unrestricted access. Its main safeguard was instructing the tutor to give teacher-designed hints instead of answers. A classroom deployment of CodeAid, an LLM programming assistant built not to reveal code solutions, with 700 students produced design principles that include avoiding direct answers while encouraging motivated learning (Kazemitabaar et al., 2024). A follow-up study compared seven cognitive engagement techniques on the friction they added, learners' ability to apply concepts to isomorphic tasks without AI, and how well learners' perceived coding ability matched their actual ability. Guiding learners step by step, prompting them to say what needed to happen at each stage before the corresponding code appeared, came out as the most effective technique (Kazemitabaar et al., 2025). Metacognitive prompts inserted into GenAI search led university students to explore more topics and ask deeper follow-up questions, with effects that depended on students' metacognitive flexibility (Singh et al., 2025).

The order of work matters as well. In a pre-registered experiment with 1,200 participants solving math practice problems, LLM-generated explanations improved learning compared with seeing only correct answers, and the largest benefits came when participants tried the problems on their own before consulting the explanations (Kumar et al., 2025). At course scale, Harvard's CS50 deployed course-specific AI tools designed to guide students toward solutions rather than hand them over, while limiting the use of commercial AI tools. Students reported that the tools felt like having a personal tutor (Liu et al., 2024). That report describes deployment and student perceptions, not measured learning or calibration. All of these designs respond to the argument that GenAI places high metacognitive demands on users and that systems should support metacognitive monitoring and control (Tankelevitch et al., 2024).

Assessment. Engagement-forcing designs are the only family with direct evidence of preserved learning, and they speak to the second reference point, the learning goal, as well as the first. Their costs are friction and lower satisfaction, which in voluntary settings may push students toward unrestricted tools instead.

5.3 Changing the costs and benefits of verification

A third view treats overreliance as a rational choice rather than a bias. Across five studies, Vasconcelos et al. (2023) showed that task difficulty, explanation difficulty, and monetary incentives changed overreliance, and argued that people strategically weigh the cost of verifying AI output against the benefit. In their account, explanations fail to reduce overreliance when they do not lower the cost of verification enough. For learners, this suggests that assignments and grading shape reliance as much as interface design does. If verification is costly and unrewarded, even a well-calibrated student may rationally lean on the AI. As far as I know, this mechanism has not been tested directly with GenAI in a course setting.

5.4 Building learner capacity

The fourth family targets the learner rather than the interface. AI literacy has been defined as a set of competencies that let people critically evaluate AI and use it effectively (Long & Magerko, 2020), and reviews have organized it around knowing and understanding, using and applying, evaluating and creating, and ethical issues (Ng et al., 2021). A systematic review of 179 documents on AI literacy in K-12 education found very few learning experiences that assessed whether students actually understood AI concepts afterward (Casal-Otero et al., 2023).

AI literacy does seem to matter as a learner characteristic. It significantly moderated the relationship between trust and appropriate reliance among undergraduates (Pitts et al., 2026b). Whether AI literacy programs produce that benefit is a separate question, and the K-12 review suggests it is rarely tested. I would argue literacy interventions should be evaluated on calibration, not only on knowledge or skepticism, because more skepticism on its own could just move students from overreliance to rejecting correct advice.

A related lever is how an AI system is framed when students first meet it. Since priming beliefs about an AI's motives changed how the same system was experienced (Pataranutaporn et al., 2023), the way a classroom AI tool is introduced, including what the instructor says about its reliability and purpose, is itself a trust intervention. It is rarely treated as one.

5.5 Teachers and institutions as part of the calibration system

Students do not calibrate their trust alone. Teachers decide which tools are allowed, how assignments reward verification, and how AI is described in class. Teachers' own trust has been studied too. In a professional development program for K-12 science teachers working with an AI grading tool, explaining how the AI makes decisions compared with human experts, and how it could complement rather than replace teachers, reduced concerns and improved trust (Nazaretsky et al., 2022). That program aimed to increase trust, not to calibrate it, which shows how the goal of adoption can differ from the goal of appropriate reliance. Institutional choices matter as well. CS50's policy of encouraging course-specific tools while limiting commercial ones (Liu et al., 2024) is in effect a decision about which system students will learn to trust. The cost-benefit account of overreliance (Vasconcelos et al., 2023) suggests that grading policies, by changing the payoff for checking AI output, may matter as much as interface design. None of these institutional levers has been evaluated on students' calibration.

5.6 Across the four families: three patterns

Reading the intervention studies together, I see three patterns that no single family shows on its own.

Adding information is not the same as adding thought. Interventions that give learners more information about the AI, such as explanations, sources, or confidence, work only if learners engage with that information. Explanations increased reliance regardless of correctness in two separate studies (Bansal et al., 2021; Kim et al., 2025), while designs that required people to think before or while seeing AI output reduced overreliance (Buçinca et al., 2021) or improved learning (Kazemitabaar et al., 2025; Kumar et al., 2025). The cost-benefit account offers one explanation: information may help only when it lowers the cost of verification enough to make checking worthwhile (Vasconcelos et al., 2023). In practice, a reliability cue attached to an answer the student can simply copy may do little, while the same cue attached to a task that requires the student's own attempt may do more.

What works is often what users like least. The designs that reduced overreliance most got the least favorable ratings (Buçinca et al., 2021), and among students, overreliance has been found to correlate with post-task trust and satisfaction (Pitts et al., 2026a). If schools choose tools by satisfaction, they may end up selecting the designs that calibrate trust worst. This tension needs direct study.

The evidence is strongest where it transfers least. The best-controlled evidence on reliability communication comes from general decision and question-answering tasks. The outcome that matters most for education, durable learning measured without AI, shows up mainly in a small number of education-specific studies of engagement-forcing designs. Very few of the studies reviewed here measure both calibration and learning in the same design. To me, that is the clearest opening in the field.

6. For Whom Do Interventions Work?

A finding that keeps coming up across this literature is that the benefits of AI assistance and of calibration interventions are not evenly spread.

Individual differences moderate intervention effects. Cognitive forcing helped people higher in need for cognition more (Buçinca et al., 2021). Among students, the relationship between trust and appropriate reliance was moderated by AI literacy and need for cognition (Pitts et al., 2026b), and metacognitive prompts worked differently depending on metacognitive flexibility (Singh et al., 2025).

Prior knowledge shapes who gains. Struggling novice programmers were the ones for whom GenAI compounded metacognitive difficulties, while students who were not struggling used it to speed up work they already understood (Prather et al., 2024). Trust itself varies by group: in a validated instrument study of 665 students, gender and educational background were significantly correlated with trust in AI-powered educational technology (Nazaretsky et al., 2025).

Inequality may begin before calibration. Who uses GenAI at all is also unequal. In a survey of 18,000 Danish workers in exposed occupations, linked to register data, women were 16 percentage points less likely than men to have used ChatGPT for work, and users had already earned slightly more before ChatGPT arrived. Workers cited employer restrictions and a perceived need for training as barriers (Humlum & Vestergaard, 2025). These are workers, not students. Still, the finding is a reminder that calibration interventions reach only people who use the tools, and that gaps in access and training may add to differences in calibration.

Work and learning may diverge. This pattern contrasts with evidence from the workplace, where GenAI has tended to compress performance differences. In a writing experiment with professionals, inequality between workers decreased with ChatGPT access (Noy & Zhang, 2023), and among customer-support agents, less experienced and lower-skilled workers gained the most (Brynjolfsson et al., 2025). One way to reconcile these, which I offer as a hypothesis and not a finding, is that AI compresses differences in output, which is what workplace studies measure, while it can widen differences in learning, because the learners who most need practice are also the most likely to substitute AI for it.

Friction may sort learners. If engagement-forcing designs are the most effective and the least liked (Buçinca et al., 2021), and if they help learners with higher need for cognition more, then in voluntary settings they may mostly reach the learners who need them least. Buçinca et al. (2021) explicitly audited their work for such "intervention-generated inequalities." Audits like that are uncommon in the education studies I reviewed, although moderation analyses such as those in Pitts et al. (2026b) point the same way. They should become standard.

Toward equitable calibration. If interventions help unevenly, one response is to adapt them. A system could add more scaffolding, such as requiring an attempt before revealing a solution, when signals suggest a student is substituting AI for their own work, and less when a student is using AI to check work they already understand. The divide Prather et al. (2024) observed, between students who used GenAI to speed up work they understood and students for whom it compounded difficulties, hints at what those signals might look like. Adaptive designs carry their own risks, including misclassifying students and applying friction unevenly, so they should be evaluated with the same subgroup analyses I recommend for fixed designs.

7. Measurement Problems in the Evidence Base

Several measurement issues limit what the current literature can conclude.

Trust level is not calibration. Some education studies measure self-reported trust or readiness to adopt AI (e.g., Amoozadeh et al., 2024; Nazaretsky et al., 2025). That is useful, but it does not show whether reliance is appropriate. Reviews of trust calibration research found that how calibration is measured often limits how interventions can be interpreted (Wischnewski et al., 2023), and that definitions of trust are often missing from study protocols (Vereschak et al., 2021). Behavioral paradigms that seed incorrect AI suggestions, as in Pitts et al. (2026a, 2026b), tell us more.

Assisted performance is not learning. Studies that measure only performance with AI present can mislead, as the contrast between the two phases of Bastani et al. (2025) shows. The most informative designs measure transfer with AI access removed (Bastani et al., 2025; Darvishi et al., 2024; Kazemitabaar et al., 2025).

Short studies dominate. A review of empirical research on trust in AI pointed to a heavy reliance on short-term, small-sample experiments, where trust develops differently than in longer-term, higher-stakes settings (Glikson & Woolley, 2020). Most intervention studies in Table 1 are single-session experiments. Semester-long deployments (Kazemitabaar et al., 2024) and field experiments (Bastani et al., 2025; Kestin et al., 2025) are still the exception. Because trust erodes quickly after confident errors and recovers slowly (Dhuliawala et al., 2023), single-session studies may misjudge how interventions perform once students have seen an AI fail.

Table 2 lists the constructs that a study of calibration in learning should keep separate, with an example measure for each.

Table 2. Constructs to measure separately in studies of learners' reliance on AI.

Construct	What it captures	Example measure	Example study
Trust	Attitude toward the system	Validated questionnaire	Nazaretsky et al. (2025)
Reliance	Behavior	Agreement with AI on seeded correct and incorrect items	Pitts et al. (2026a, 2026b)
Appropriateness of reliance	Whether behavior fits AI correctness	Two-dimensional appropriateness-of-reliance measure	Schemmer et al. (2023)
Calibration and discrimination	Fit between user confidence and AI accuracy	Calibration gap; ability to separate correct from incorrect answers	Steyvers et al. (2025)
Metacognitive monitoring	Accuracy of self-assessment	Gap between perceived and actual ability	Kazemitabaar et al. (2025)
Learning	Durable skill	Transfer test with AI access removed	Bastani et al. (2025); Kazemitabaar et al. (2025)
Distribution of effects	Who benefits	Effects by prior knowledge, AI literacy, need for cognition	Buçinca et al. (2021); Pitts et al. (2026b)

8. Implications for Practice

The evidence base is still developing, but I think several implications for educators, designers, and institutions follow from it. I offer them as reasonable inferences, not tested guidelines.

For designers of educational AI. Be careful with explanations, since they can raise reliance on wrong answers as much as on right ones (Bansal et al., 2021; Kim et al., 2025). Prefer designs that ask for the learner's own thinking before revealing a solution (Kazemitabaar et al., 2025; Kumar et al., 2025), and build in safeguards for learning, as in the guarded tutor of Bastani et al. (2025). Where the system communicates reliability, use first-person uncertainty and point to sources (Kim et al., 2024, 2025), and check whether the model expresses uncertainty at all, since many do not by default (Zhou et al., 2024).

For instructors. For each assignment, decide which parts are the learning target, and tell students. That makes the second reference point in Figure 1 explicit instead of leaving students to guess it. Think about how AI tools are introduced, since framing shapes trust (Pataranutaporn et al., 2023), and consider assessment designs that reward checking AI output (Vasconcelos et al., 2023). Keep some assessment AI-free, because assisted performance can hide lost learning (Bastani et al., 2025).

For institutions. Evaluate AI tools on students' calibration and learning, not only on satisfaction or usage, which may favor the designs that calibrate trust least (Buçinca et al., 2021; Pitts et al., 2026a). Pay

attention to who is not using the tools and who is using them without guidance, since both access and benefit appear to be unevenly distributed (Humlum & Vestergaard, 2025; Prather et al., 2024).

9. Future Directions

1. **Test decision-support interventions with learners.** The best-studied reliability interventions, including uncertainty expressions, calibrated explanations, sources, and contrastive information, have rarely been tested with students on learning tasks. Replications that add delayed, AI-free transfer measures would show whether they calibrate only assisted performance or also support learning.
2. **Design for both reference points.** Interventions should be judged on whether they help learners accept correct output, reject incorrect output, and do the work that is the learning target. Designs that adapt what the AI reveals to the learning goal, such as guarded tutors and step-by-step engagement, are promising starting points.
3. **Report calibration, not only trust.** Studies should seed incorrect AI output, report appropriate reliance on both correct and incorrect cases (Schemmer et al., 2023), and distinguish calibration from discrimination (Steyvers et al., 2025).
4. **Audit for intervention-generated inequalities.** Every intervention study should report effects separately by prior knowledge, AI literacy, and need for cognition at minimum, following the audit practice of Buçinca et al. (2021). Where effects differ, adaptive designs that give more scaffolding to learners at risk of substitution should be tested.
5. **Evaluate AI literacy on calibration.** Few K-12 AI literacy programs assess what students learned (Casal-Otero et al., 2023), yet AI literacy moderates how trust turns into reliance (Pitts et al., 2026b). Programs should be tested on whether they improve appropriate reliance on both correct and incorrect AI output, not only on knowledge or skepticism.
6. **Design tutors that express uncertainty.** Given that LLMs rarely express uncertainty and are overconfident when they do (Zhou et al., 2024), educational AI should be tested on whether it communicates calibrated uncertainty. First-person uncertainty expressions reduced overreliance in a medical question-answering task (Kim et al., 2024). Whether they work the same way for learners, and whether they affect motivation, is untested.
7. **Treat introductions and policies as interventions.** How instructors and institutions introduce an AI tool, and how grading rewards verification, plausibly shape students' trust (Pataranutaporn et al., 2023; Vasconcelos et al., 2023). These cheap levers should be evaluated with the same calibration measures as interface designs.
8. **Study longer horizons and real courses.** Trust has a learned layer alongside its dispositional and situational layers (Hoff & Bashir, 2015). Semester-scale studies that track how students' calibration changes as they run into AI errors would address a gap that runs through this whole literature.

10. Conclusion

The studies I reviewed point to one main message. The educational problem is not that students trust generative AI too much or too little. It is that their trust is often poorly calibrated, and fluent explanations can make it worse. Interventions that communicate reliability can reduce overreliance on incorrect

answers, but most were tested outside learning tasks, and none of those studies measured learning with the AI removed. Interventions that require learners to think before accepting are the only family with direct evidence of preserved learning, yet people like them less, and they tend to help the learners who were already best prepared. GenAI makes all of this harder, because the systems themselves rarely signal doubt. Calibrating learners' trust will take designs that treat the learning goal as seriously as the AI's accuracy, and research that asks of every intervention not only whether it works but for whom.

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